NLP Group Project

Hotel Review Sentiment

Classification

Building a Binary Sentiment Classifier for TripAdvisor Reviews

Raw Text -> Preprocessing -> Tokenization -> Encoding -> Padding -> Model -> Prediction

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Introduction to AI — Basic Lab

Project Goal & Approach

Goal:Our goal is building a sentiment classifier that is binary for real-life reviews for hotels. Choose a bag of words algorithm and logistic regression, check the results and demonstrate its limitation with word order.

Approach 1 — Baseline

- Bag-of-Words (BoW) representation
- 3,000 most frequent words as features
- Logistic Regression classifier
- Fast and interpretable

Approach 2 — Sequential

- Recurrent Neural Network (RNN)
- Reads one word at a time
- Maintains a hidden state (memory)
- Handles negations & word order

Classify

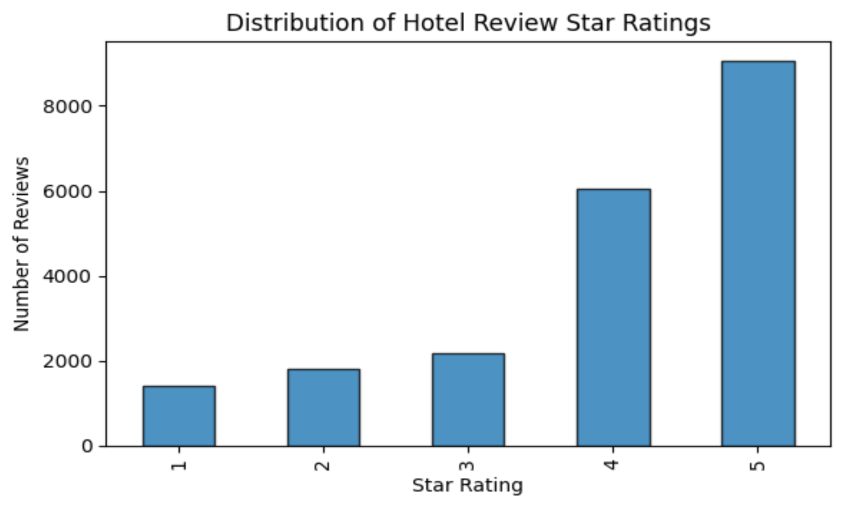
Dataset: TripAdvisor Hotel Reviews

20,491
Total Reviews

1 – 5
Star Ratings

Real
User-Generated

Rating Distribution



Why This Dataset?

- Exposes BoW word-order limitations
- Real and user-generated language
- Full of negations
- Stars provide clear ground truth
- 20k+ reviews — large enough for ML

From 5-Star Ratings to Binary Labels

Rating 4 – 5 Stars

Label them as positive

15,093

82.4%

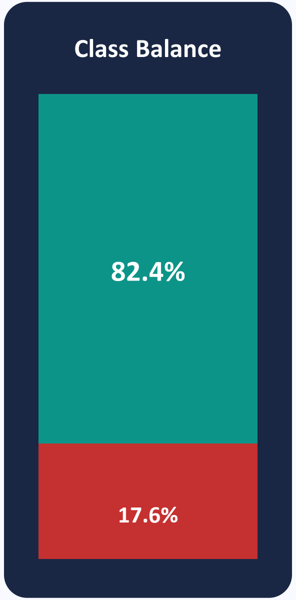
Rating 1 – 2 Stars

Label them as negative.

3,214

17.6%

3 stars are dropped because they can mean neutral.



Data Split & Preprocessing

TRAINING	VALIDATION	TEST
12,814	2,746	2,747
70%	15%	15%

Text Preprocessing (Zhan Shakarim)

1. **Lowercase:** "The Room was SPOTLESS!!!" -> "the room was spotless"
2. **Remove non-letters:** Punctuation, numbers, symbols stripped with regex [^a-z\s]
3. **Collapse spaces:** "Never   going   back" -> "never going back"

Tokenization, Vocabulary & Padding

56,030
Vocabulary Size

101.5
Avg Review Length (tokens)

266
Max Padded Length (95th pct)

Special Tokens

<PAD> (index 0): Pad for empty spaces
<UNK> (index 1): Unknown words

Padding

We padded the reviews to 266 tokens, which is inside the 95th percent of training lengths. That way we can avoid the outliers and also, we will have a more stable model.

Encoding Example

```
"perfect friend told hotel ..."  ->  [15475, 54656, 40756, 21829, ...]

Decode: 15475 -> 'perfect',  54656 -> 'friend',  40756 -> 'told'  ...
```

Bag-of-Words + Logistic Regression (Nebi Goncu)

Bag-of-Words (BoW)

- Each review = vector of word counts
- 3,000 most frequent words as features
- Word order and grammar ignore
- Feature matrix: 12,814 x 3,000

Logistic Regression

- Linear classifier on BoW features
- Outputs probability per class
- Class weights balance 82/18 split
- max_iter=1000 for convergence

Live Predictions

Amazing staff and the room was spotless.	Positive	91.4%
Rooms were gross and staff were rude.	Negative	88.9%
One loved everything, calling it perfect.	Positive	93.5%
Location good, service bad; too expensive.	Negative	83.3%
It's a mixed bag, really.	Positive	53.2%

Results: BoW + Logistic Regression

94.76%

Validation Accuracy

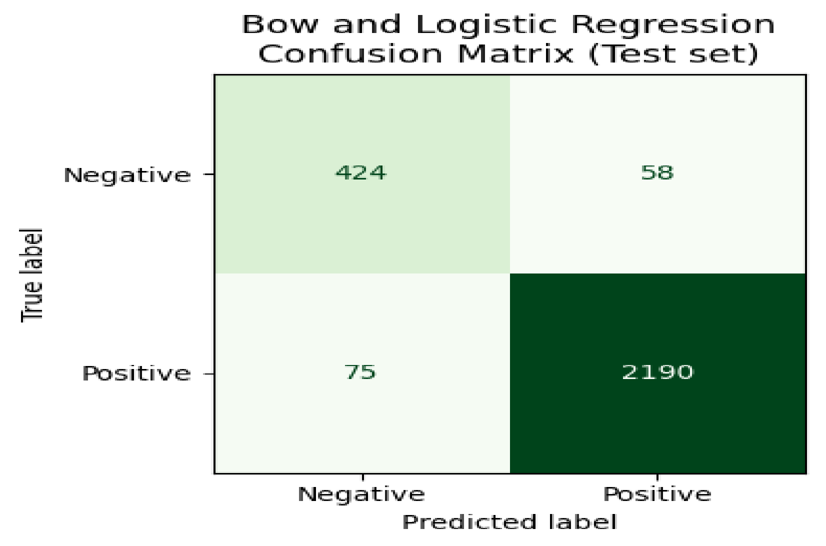
2,746 reviews

95.16%

Test Accuracy

2,747 reviews

Confusion Matrix (Test Set)



Classification Report

Class	Precision	Recall	F1
Negative	0.85	0.88	0.86
Positive	0.97	0.97	0.97
Macro avg	0.91	0.92	0.92

What the Model Learned: Top Sentiment Words

Top 10 POSITIVE Words	Top 10 NEGATIVE Words
1. loved	1. poor
2. perfect	2. disappointing
3. fantastic	3. worst
4. fabulous	4. shabby
5. pleased	5. terrible
6. complaint	6. poorly
7. spectacular	7. overpriced
8. spotless	8. cleaner
9. outstanding	9. sadly
10. london	10. horrible

The Key Limitation: Word Order Blindness

BOW sees a bag of words that is unordered, so the word order is totally invisible to the model.

"staff were never rude to us"

Correct reading: staff were NOT rude (positive intent)

Negative

78.04%

Same words — same prediction. BoW cannot distinguish!

"staff were rude never to us"

Grammatically odd — but exact same prediction!

Negative

78.04%

Solution: An RNN reads words sequentially with memory — it CAN distinguish 'never rude' from 'rude never'.

Summary

BoW + LogReg — Strengths

- Simple and fast — no deep learning required
- 95% test accuracy on hotel sentiment
- Highly interpretable learned word weights
- Excellent baseline for any NLP task

BoW + LogReg — Limitations

- Reviews treated as unordered word bags
- Cannot understand negations (never rude vs rude)
- Word order and grammar invisible
- Fails on subtle, context-dependent meaning

For a better result, especially with negations and word orders, a sequential model like RNN is needed.